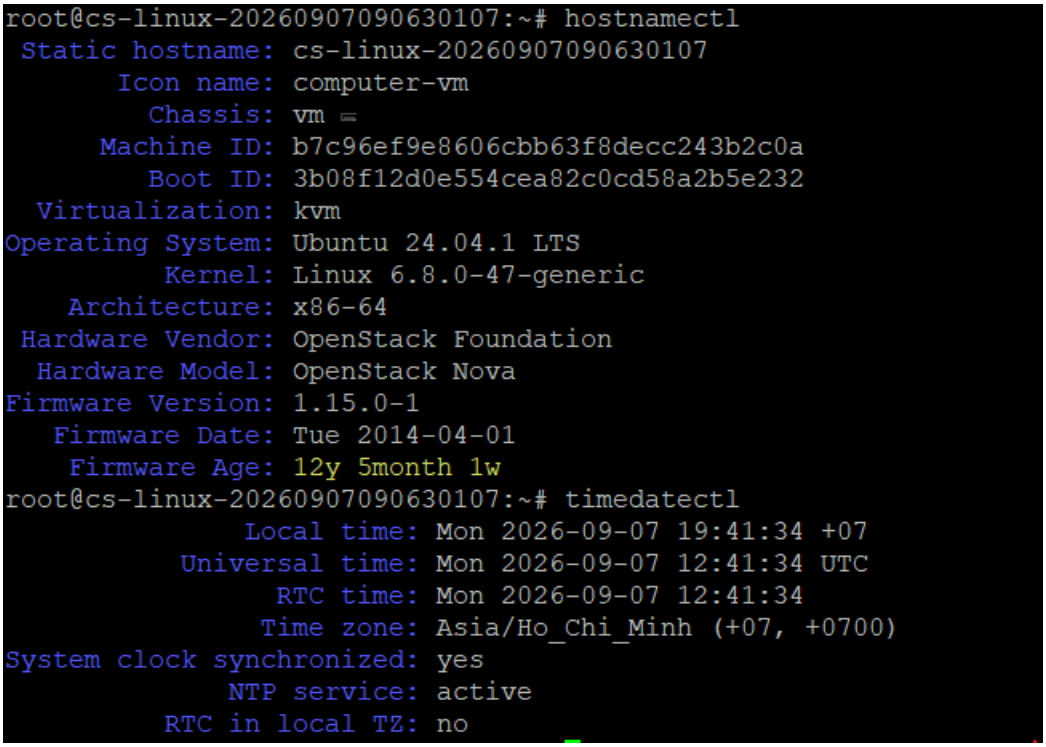
BÁO CÁO BỘ 5 BÀI TẬP LINUX

BÀI 1:

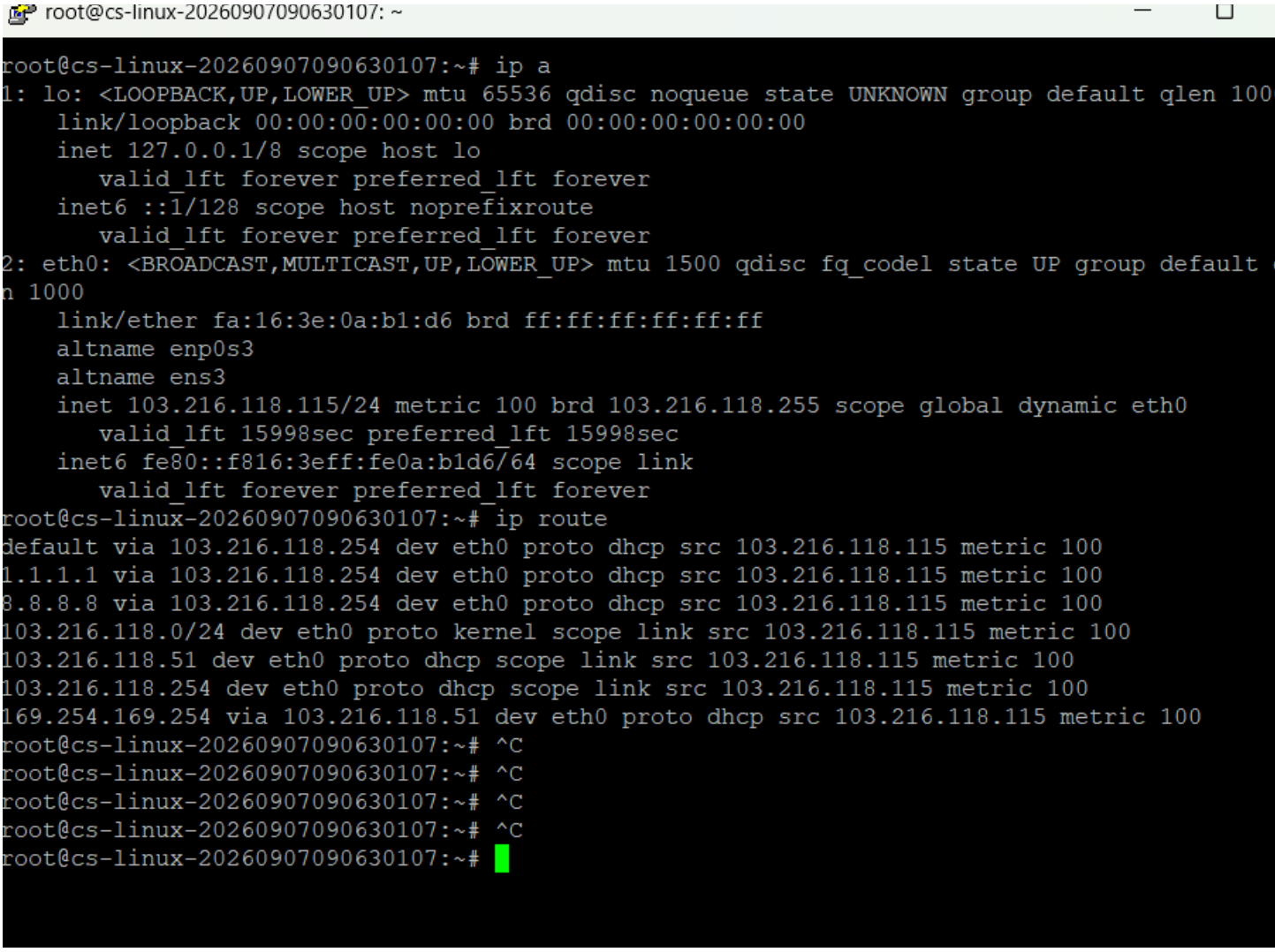
1 . Xác định phiên bản Ubuntu, kernel, hostname, múi giờ, thời gian hệ thống và tình trạng đồng bộ thời gian.

Minh chứng:



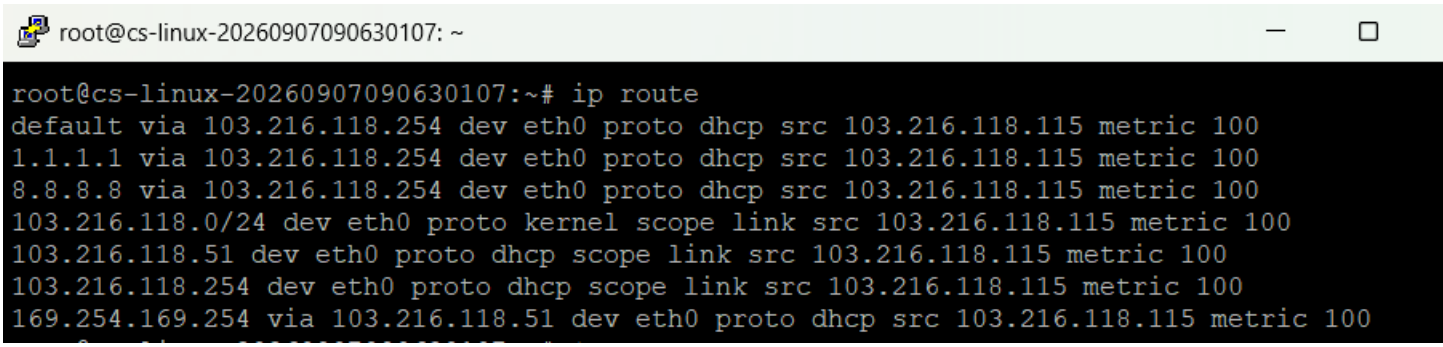
2. Xác định toàn bộ network interface, trạng thái UP/DOWN, địa chỉ IPv4/IPv6, MTU và interface đang mang default route

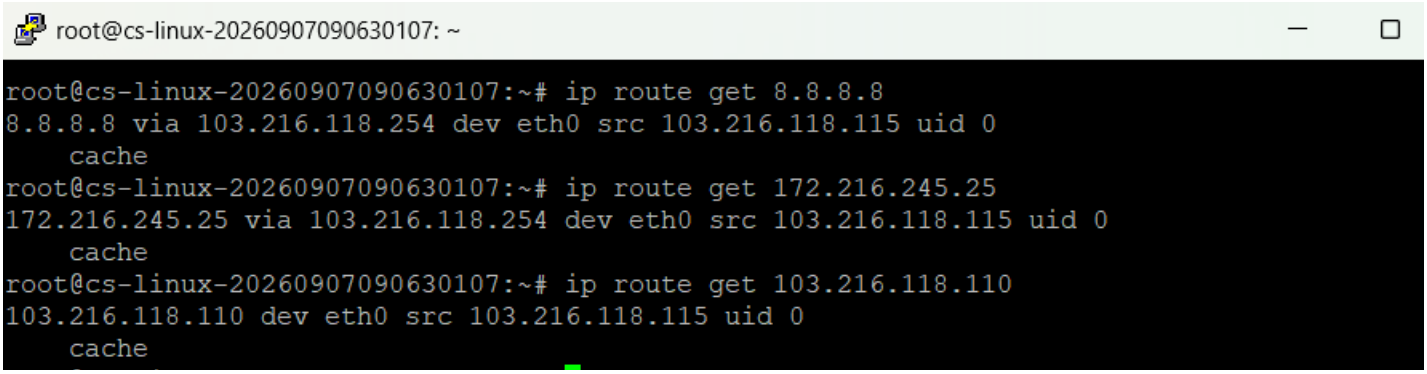
Minh chứng



3. Lập bảng route hiện tại, default gateway và đường đi dự kiến của lưu lượng đến Internet, VPN và mạng nội bộ.

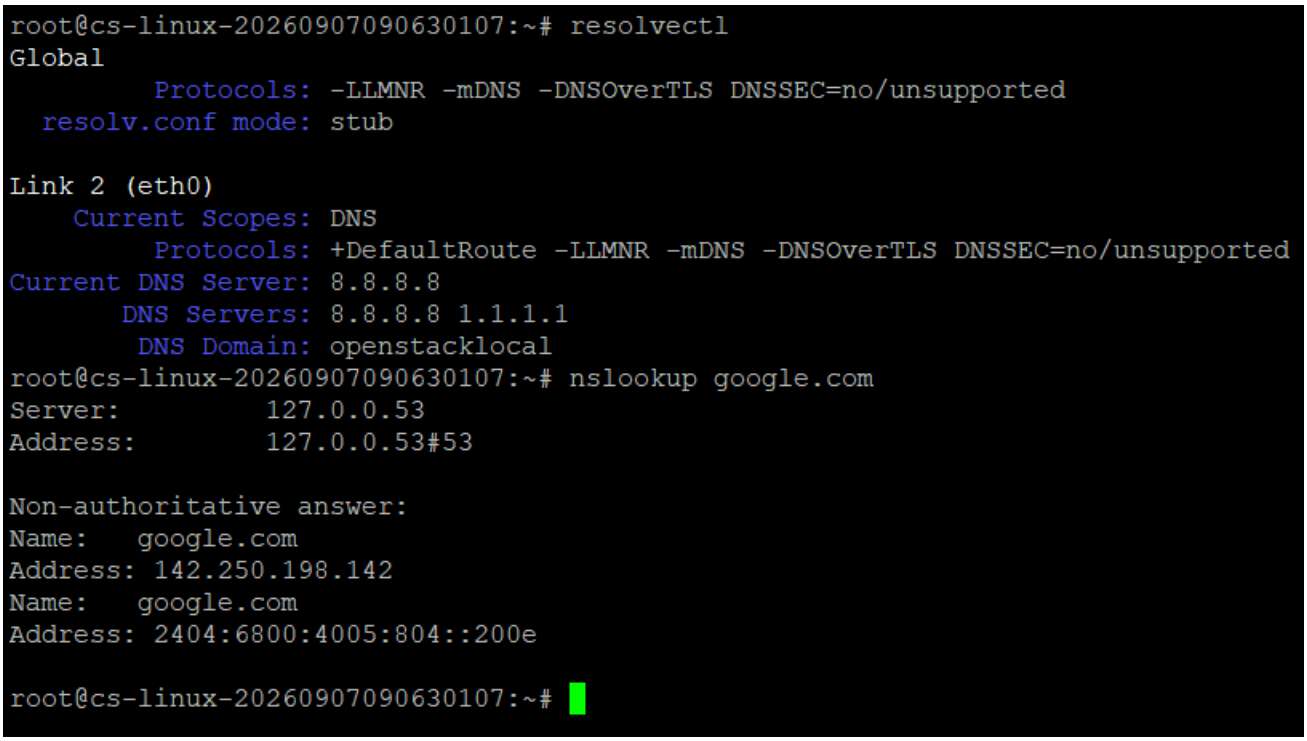
Minh chứng :





4. Xác định DNS server đang được sử dụng; kiểm tra phân giải tên miền và phân biệt lỗi DNS với lỗi mất kết nối mạng.

Minh chứng



B: Phân biệt lỗi DNS với lỗi mất kết nối mạng

Tình huống	ping 8.8.8.8	ping google.com	Kết luận khoanh vùng
Mạng OK, DNS OK	Thành công	Thành công	Hệ thống bình thường
Mạng OK, DNS hỏng	Thành công	Thất bại	Lỗi nằm ở tầng phân giải tên, không phải kết nối
Mất kết nối mạng	Thất bại	Thất bại	Lỗi kết nối, DNS chỉ là nạn nhân

5. Thống kê toàn bộ cổng TCP/UDP đang lắng nghe; với mỗi cổng phải chỉ ra tiến trình, PID, user chạy tiến trình và dịch vụ liên quan.

Minh chứng:

```
root@cs-linux-20260907090630107:~# ss -tulnpe
Netid  State   Recv-Q  Send-Q                     Local Address:Port     Peer Address:Por
Process
udp    UNCONN  0       0                             127.0.0.54:53           0.0.0.0:*
 users:(("systemd-resolve",pid=37890,fd=16)) uid:992 ino:102297 sk:1 cgroup:/system.slice/
temd-resolved.service <->
udp    UNCONN  0       0                          127.0.0.53%lo:53           0.0.0.0:*
 users:(("systemd-resolve",pid=37890,fd=14)) uid:992 ino:102294 sk:2 cgroup:/system.slice/
temd-resolved.service <->
udp    UNCONN  0       0                    103.216.118.115%eth0:68           0.0.0.0:*
 users:(("systemd-network",pid=37922,fd=24)) uid:998 ino:169486 sk:3 cgroup:/system.slice/
temd-networkd.service <->
udp    UNCONN  0       0                           127.0.0.1:323           0.0.0.0:*
 users:(("chronyd",pid=57451,fd=5)) ino:144972 sk:4 cgroup:/system.slice/chrony.service <-
udp    UNCONN  0       0        [fe80::f816:3eff:fe0a:b1d6]%eth0:546             [::]:*
 users:(("systemd-network",pid=37922,fd=23)) uid:998 ino:102633 sk:5 cgroup:/system.slice/
temd-networkd.service v6only:1 <->
udp    UNCONN  0       0                                [::1]:323             [::]:*
 users:(("chronyd",pid=57451,fd=6)) ino:144973 sk:6 cgroup:/system.slice/chrony.service v6
y:1 <->
tcp    LISTEN  0       4096                         127.0.0.53%lo:53           0.0.0.0:*
 users:(("systemd-resolve",pid=37890,fd=15)) uid:992 ino:102296 sk:7 cgroup:/system.slice/
temd-resolved.service <->
tcp    LISTEN  0       4096                          127.0.0.54:53           0.0.0.0:*
 users:(("systemd-resolve",pid=37890,fd=17)) uid:992 ino:102298 sk:8 cgroup:/system.slice/
temd-resolved.service <->
tcp    LISTEN  0       4096                            0.0.0.0:24700         0.0.0.0:*
 users:(("sshd",pid=57789,fd=3),("systemd",pid=1,fd=203)) ino:100013 sk:9 cgroup:/system.s
e/ssh.socket <->
tcp    LISTEN  0       4096                              [::]:24700             [::]:*
 users:(("sshd",pid=57789,fd=4),("systemd",pid=1,fd=204)) ino:100017 sk:a cgroup:/system.s
e/ssh.socket v6only:1 <->
```

6. Lập bảng phân loại cổng: bắt buộc phải giữ, được phép đóng, chưa xác định mục đích và cần giảng viên xác nhận.

7. Kiểm tra các service đang chạy, service lỗi, service tự khởi động cùng hệ thống

```
root@maychucuavietanh:~# systemctl list-units --type=service --state=running --no-pager
  UNIT                          LOAD   ACTIVE SUB     DESCRIPTION
  chrony.service                loaded active running chrony, an NTP client/server
  cron.service                  loaded active running Regular background program processing
  dbus.service                  loaded active running D-Bus System Message Bus
  getty@tty1.service            loaded active running Getty on tty1
  ModemManager.service          loaded active running Modem Manager
  multipathd.service            loaded active running Device-Mapper Multipath Device Control
  networkd-dispatcher.service   loaded active running Dispatcher daemon for systemd-networkd
  polkit.service                loaded active running Authorization Manager
  qemu-guest-agent.service      loaded active running QEMU Guest Agent
  rsyslog.service               loaded active running System Logging Service
  serial-getty@ttyS0.service    loaded active running Serial Getty on ttyS0
  ssh.service                   loaded active running OpenBSD Secure Shell server
  systemd-journald.service      loaded active running Journal Service
  systemd-logind.service        loaded active running User Login Management
  systemd-networkd.service      loaded active running Network Configuration
  systemd-resolved.service      loaded active running Network Name Resolution
  systemd-udevd.service         loaded active running Rule-based Manager for Device Events a
  tuned.service                 loaded active running Dynamic System Tuning Daemon
  udisks2.service               loaded active running Disk Manager
  unattended-upgrades.service   loaded active running Unattended Upgrades Shutdown
  upower.service                loaded active running Daemon for power management
  user@0.service                loaded active running User Manager for UID 0

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

22 loaded units listed.
```

```
root@maychucuavietanh:~# systemctl --failed
  UNIT LOAD ACTIVE SUB DESCRIPTION

0 loaded units listed.
```

```
UNIT FILE                                  STATE   PRESET
apparmor.service                           enabled enabled
apport.service                             enabled enabled
blk-availability.service                   enabled enabled
chrony.service                             enabled enabled
cloud-config.service                       enabled enabled
cloud-final.service                        enabled enabled
cloud-init-local.service                   enabled enabled
cloud-init.service                         enabled enabled
console-setup.service                      enabled enabled
cron.service                               enabled enabled
dmesg.service                              enabled enabled
e2scrub_reap.service                       enabled enabled
finalrd.service                            enabled enabled
getty@.service                             enabled enabled
gpu-manager.service                        enabled enabled
grub-common.service                        enabled enabled
grub-initrd-fallback.service               enabled enabled
keyboard-setup.service                     enabled enabled
lvm2-monitor.service                       enabled enabled
ModemManager.service                       enabled enabled
multipathd.service                         enabled enabled
networkd-dispatcher.service                enabled enabled
open-iscsi.service                         enabled enabled
open-vm-tools.service                      enabled enabled
pollinate.service                          enabled enabled
rsyslog.service                            enabled enabled
secureboot-db.service                      enabled enabled
setvtrgb.service                           enabled enabled
snapd.apparmor.service                     enabled enabled
snapd.autoimport.service                   enabled enabled
snapd.core-fixup.service                   enabled enabled
snapd.recovery-chooser-trigger.service     enabled enabled
snapd.seeded.service                       enabled enabled
snapd.service                              enabled enabled
snapd.system-shutdown.service              enabled enabled
sysstat.service                            enabled enabled
systemd-networkd-wait-online.service       enabled enabled
systemd-networkd.service                   enabled enabled
systemd-pstore.service                     enabled enabled
systemd-resolved.service                   enabled enabled
thermald.service                           enabled enabled
tuned.service                              enabled enabled
ua-reboot-cmds.service                     enabled enabled
ubuntu-advantage.service                   enabled enabled
udisks2.service                            enabled enabled
unattended-upgrades.service                enabled enabled
vgauth.service                             enabled enabled
```

```
root@maychucuavietanh:~# systemctl list-units  --type=socket  --state=active  --no-pager
  UNIT                                LOAD   ACTIVE SUB       DESCRIPTION
  cloud-init-hotplugd.socket          loaded active listening cloud-init hotplug hook socket
  dbus.socket                         loaded active running   D-Bus System Message Bus Socket
  dm-event.socket                     loaded active listening Device-mapper event daemon FIFOs
  iscsid.socket                       loaded active listening Open-iSCSI iscsid Socket
  lvm2-lvmpolld.socket                loaded active listening LVM2 poll daemon socket
  lxd-installer.socket                loaded active listening Helper to install lxd snap on de
  multipathd.socket                   loaded active running   multipathd control socket
  snapd.socket                        loaded active listening Socket activation for snappy dae
  ssh.socket                          loaded active running   OpenBSD Secure Shell server sock
  syslog.socket                       loaded active running   Syslog Socket
  systemd-fsckd.socket                loaded active listening fsck to fsckd communication Sock
  systemd-initctl.socket              loaded active listening initctl Compatibility Named Pipe
  systemd-journald-dev-log.socket     loaded active running   Journal Socket (/dev/log)
  systemd-journald.socket             loaded active running   Journal Socket
  systemd-networkd.socket             loaded active running   Network Service Netlink Socket
  systemd-rfkill.socket               loaded active listening Load/Save RF Kill Switch Status
  systemd-sysext.socket               loaded active listening System Extension Image Managemen
  systemd-udevd-control.socket        loaded active running   udev Control Socket
  systemd-udevd-kernel.socket         loaded active running   udev Kernel Socket
  uuidd.socket                        loaded active listening UUID daemon activation socket

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.
```

8. Kiểm tra CPU, RAM, swap, dung lượng từng filesystem, inode và 10 thư mục hoặc file chiếm nhiều dung lượng nhất trong phạm vi được phép.

```
root@maychucuavietanh:~# df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           197M  1.1M  196M   1% /run
/dev/sda2        20G  3.5G   17G  18% /
tmpfs           985M     0  985M   0% /dev/shm
tmpfs           5.0M     0  5.0M   0% /run/lock
tmpfs           197M   12K  197M   1% /run/user/0
root@maychucuavietanh:~# df -i
Filesystem       Inodes  IUsed    IFree IUse% Mounted on
tmpfs            251921    698   251223    1% /run
/dev/sda2      10484720 134797 10349923    2% /
tmpfs            251921      1   251920    1% /dev/shm
tmpfs            251921      3   251918    1% /run/lock
tmpfs             50384     33    50351    1% /run/user/0
root@maychucuavietanh:~# du -ah max-depth=2
du: cannot access 'max-depth=2': No such file or directory
root@maychucuavietanh:~# du -ah --max-depth=2 | sort -hr | head -n 10
24K     .
8.0K    ./.bash_history
4.0K    ./.ssh/authorized_keys
4.0K    ./.ssh
4.0K    ./.profile
4.0K    ./.lesshst
4.0K    ./.bashrc
0       ./.local/share
0       ./.local
0       ./.config/htop
root@maychucuavietanh:~# lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          40 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   1
  On-line CPU(s) list:    0
Vendor ID:                GenuineIntel
  BIOS Vendor ID:         QEMU
  Model name:             Intel Core Processor (Broadwell, no TSX, IBRS)
    BIOS Model name:      pc-i440fx-6.2  CPU @ 2.0GHz
    BIOS CPU family:      1
    CPU family:           6
    Model:                61
    Thread(s) per core:   1
    Core(s) per socket:   1
    Socket(s):            1
    Stepping:             2
    BogoMIPS:             4399.99
```

9. Kiểm tra lịch sử đăng nhập, người dùng đang online và các phiên SSH hiện tại

```
root@maychucuavietanh:~# last
root     pts/0        14.0.22.154      Wed Sep  9 08:11   still logged in
root     pts/0        14.0.22.154      Wed Sep  9 07:53 - 08:02  (00:08)
root     pts/1        138.199.62.8     Tue Sep  8 23:37 - 23:48  (00:10)
root     pts/0        27.79.217.108    Tue Sep  8 23:34 - 01:48  (02:13)
root     pts/1        27.79.217.108    Tue Sep  8 22:43 - 22:53  (00:10)
root     pts/0        27.79.217.108    Tue Sep  8 22:24 - 23:31  (01:07)
root     pts/0        42.113.162.76    Tue Sep  8 19:37 - 22:06  (02:28)
root     pts/0        14.0.22.154      Tue Sep  8 15:03 - 15:48  (00:44)
root     pts/0        14.0.22.154      Tue Sep  8 14:19 - 14:23  (00:03)
root     pts/3        103.238.69.131   Tue Sep  8 09:00 - 11:14  (02:14)
root     pts/3        42.113.162.76    Tue Sep  8 07:14 - 07:24  (00:09)
root     pts/2        172.216.245.15   Tue Sep  8 07:09 - 09:14  (02:05)
root     pts/2        42.113.162.76    Tue Sep  8 07:08 - 07:08  (00:00)
root     pts/1        172.216.245.25   Tue Sep  8 07:07 - 09:10  (02:03)
root     pts/0        42.113.162.76    Tue Sep  8 07:03 - 09:17  (02:13)
root     pts/0        42.113.162.76    Tue Sep  8 05:42 - 06:44  (01:02)
root     pts/0        42.113.162.76    Mon Sep  7 19:39 - 20:19  (00:40)
root     pts/3        14.0.22.154      Mon Sep  7 15:43 - 15:50  (00:07)
root     pts/2        146.70.192.172   Mon Sep  7 15:38 - 17:38  (02:00)
root     pts/1        14.0.22.154      Mon Sep  7 14:30 - 17:41  (03:11)
root     pts/1        14.0.22.154      Mon Sep  7 13:57 - 14:01  (00:03)
root     pts/0        14.0.22.154      Mon Sep  7 13:18 - 15:51  (02:33)
reboot   system boot  6.8.0-47-generic Mon Sep  7 09:07   still running

wtmp begins Mon Sep  7 09:07:02 2026
```

```
root@maychucuavietanh:~# lastb
root     ssh:notty    14.0.22.154      Wed Sep  9 08:11 - 08:11  (00:00)
root     ssh:notty    14.0.22.154      Wed Sep  9 08:10 - 08:10  (00:00)
root     ssh:notty    14.0.22.154      Wed Sep  9 08:10 - 08:10  (00:00)
root     ssh:notty    14.0.22.154      Wed Sep  9 08:10 - 08:10  (00:00)
root     ssh:notty    14.0.22.154      Wed Sep  9 08:10 - 08:10  (00:00)
root     ssh:notty    103.186.62.23    Wed Sep  9 07:20 - 07:20  (00:00)
root     ssh:notty    103.186.62.23    Wed Sep  9 05:26 - 05:26  (00:00)
root     ssh:notty    103.186.62.23    Wed Sep  9 05:04 - 05:04  (00:00)
root     ssh:notty    103.186.62.23    Wed Sep  9 02:54 - 02:54  (00:00)
root     ssh:notty    103.186.62.23    Wed Sep  9 01:22 - 01:22  (00:00)
root     ssh:notty    103.186.62.23    Wed Sep  9 00:25 - 00:25  (00:00)
root     ssh:notty    103.186.62.23    Tue Sep  8 21:09 - 21:09  (00:00)
root     ssh:notty    103.186.62.23    Tue Sep  8 19:49 - 19:49  (00:00)
root     ssh:notty    103.186.62.23    Tue Sep  8 16:40 - 16:40  (00:00)
root     ssh:notty    14.0.22.154      Tue Sep  8 15:03 - 15:03  (00:00)
root     ssh:notty    14.0.22.154      Tue Sep  8 15:03 - 15:03  (00:00)
root     ssh:notty    14.0.22.154      Tue Sep  8 15:02 - 15:02  (00:00)
root     ssh:notty    103.186.62.23    Tue Sep  8 13:56 - 13:56  (00:00)
root     ssh:notty    103.186.62.23    Tue Sep  8 11:37 - 11:37  (00:00)
root     ssh:notty    146.70.192.172   Mon Sep  7 15:37 - 15:37  (00:00)
root     ssh:notty    146.70.192.172   Mon Sep  7 15:37 - 15:37  (00:00)
root     ssh:notty    146.70.192.172   Mon Sep  7 15:37 - 15:37  (00:00)
root     ssh:notty    146.70.192.172   Mon Sep  7 15:36 - 15:36  (00:00)
root     ssh:notty    146.70.192.172   Mon Sep  7 15:35 - 15:35  (00:00)
root     ssh:notty    146.70.192.172   Mon Sep  7 15:35 - 15:35  (00:00)
root     ssh:notty    146.70.192.172   Mon Sep  7 15:35 - 15:35  (00:00)
root     ssh:notty    14.0.22.154      Mon Sep  7 13:57 - 13:57  (00:00)
root     ssh:notty    14.0.22.154      Mon Sep  7 13:57 - 13:57  (00:00)
root     ssh:notty    14.0.22.154      Mon Sep  7 13:56 - 13:56  (00:00)
root     ssh:notty    14.0.22.154      Mon Sep  7 13:56 - 13:56  (00:00)
root     ssh:notty    14.0.22.154      Mon Sep  7 13:56 - 13:56  (00:00)
root     ssh:notty    14.0.22.154      Mon Sep  7 13:56 - 13:56  (00:00)
root     ssh:notty    14.0.22.154      Mon Sep  7 13:56 - 13:56  (00:00)
root     ssh:notty    14.0.22.154      Mon Sep  7 13:18 - 13:18  (00:00)
```

```
root@maychucuavietanh:~# ss -tuln | grep 24700
tcp   LISTEN 0      4096                         0.0.0.0:24700      0.0.0.0:*
tcp   LISTEN 0      4096                            [::]:24700         [::]:*
```